**“Sensibly used, mathematical models are no more and no less than tools for thinking about things in a precise way.”**

**Roy Anderson and Robert May**

# Matching the real world

Anderson and May needed to show that their model was an accurate predictor of a real-life population. They did so by using data from a study of laboratory mice infected with the bacterial disease pasteurellosis; the data included the impact on the population of adding individuals at different rates. The observed data confirmed their predictions, so the two scientists were able to consider the effects of hypothetical values. They found, for instance, that when the rate of added mice was highest, the disease had the greatest impact on population numbers. This suggests that species with high reproductive rates (introducing large numbers of uninfected offspring) are most likely to have endemic diseases within the population, and show depressed numbers compared with species that breed more slowly. They also explored the differing effects on populations of diseases of different intensities.

Unlike endemic diseases, in which the population’s level of infection remains consistent, epidemics appear in populations when the growth rate of all infected and uninfected members is low compared to the death rate caused by the disease. Infection numbers rise sharply to a maximum, then drop away. Epidemics also occur when a disease is not particularly deadly but slows the population growth rate; this has occurred with human diseases such as measles and chickenpox.

**“Diseases such as measles and rubella, with short infections and lasting immunity, will tend to exhibit epidemic patterns.”**

**Roy Anderson**